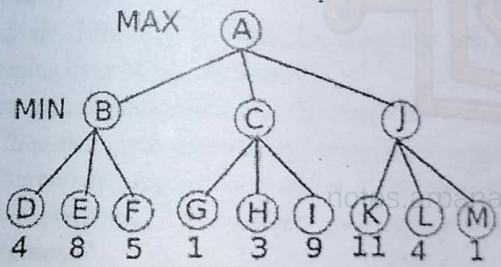
POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2014
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. Define and describe the difference between knowledge, belief, hypothesis and data. 8
2. What is the goal of a constraint solver? How are constraints propagated in a constraint satisfaction search? 7
3. Model the tic-tac-toe problem as a production system. 8
4. What is A* search? Explain with its algorithms. 7
5. In the figure below, use minimax search with alpha-beta pruning to decide the next move (node) by the MAX player from node A. The numbers indicate utility values. 8



6. Compare Breadth First Search and Depth First Search in terms of time and space complexity, completeness and optimality. 7
7. Convert (((A AND B) OR (C AND D)) OR E) into Conjunctive Normal Form. Here, OR and AND denote the usual propositional logic operators. 7
8. Assume the following facts: 8
   - Steve only likes easy courses
   - Science courses are hard

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- All the course in the basket weaving department are easy
- BK301 is a basket weaving course

Use resolution to answer the question," What course would Steve like?"

5. a) Define proposition and a predicate. How does first order logic exten propositional logic to overcome its limitations?
   b) What is a semantic net? For the statement, "The dog bit the ma carrier." Now, represent this statement (Knowledge) using partitione semantic nets.
6. a) What is learning? Explain rote learning and induction learning. Wh the process of knowledge acquisition so tedious.
   b) Why Natural Language understanding is difficult? And also expla its application.
7. Write short notes on: **(Any two)**
   a) Statistical reasoning
   b) Expert System
   c) Mathematics and Artificial Intelligence.

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